

A2 Assessment Instructions

This assessment focusses on your ability to synthesize and critique scientific ideas from scientific sources and write about them in an engaging way or the educated public. Checkpoint 3 and checkpoint 4 help practise your skills for this assessment.

Assessment criteria: Critical Analysis and Language Use

A2 includes **part 1** (written blog post) and **part 2** (video reflection on your blog post).

Part 1: an individual written scientific blog post

Scenario: You are invited to participate in a science communication project co-hosted by the School of Science and an international organisation. You should produce a written scientific blog post which **explores one scientific topic of your choice**, to **engage the reader**. It should **synthesize information from multiple sources** and **critique two perspectives on your topic**.

Purpose: inform, interest and inspire the educated public

Length: 1200-1500 words excluding reference list

References: 4-10 references. Use IEEE for in-text citation and reference list.

Style: engaging for the reader, not too formal or informal

Required content: title, subheadings, introduction, brief explanation of a key scientific concept and why it matters, in-depth critique of two perspectives around the concept, conclusion. Visuals are optional.

Stage		Advice for developing your scientific blog post
1.		Select a broad topic that interests you. To help develop ideas, read around different popular science topics in Scientific American , Nature , Popular Science , and other science blogs . Your topic must be: - scientific (not social science, psychology or pseudo science) - specific e.g. quantum computing (not vague e.g. computer science)
2.		Narrow down this broad topic to a more specific topic* that you can explore in-depth through your blog post. To help you, try reading around relevant issues or different theories on the topic and decide what remains critical, unresolved, evolving or controversial around some of ideas.
3.		Identify two perspectives* on your topic and consider how they can be meaningfully compared. Perspectives might be ONE of the following: - two different theories that could explain one phenomenon - two different interpretations or conclusions about the available evidence - two different time periods for evaluating impact (short term vs long term) - two different approaches to finding a solution - or various others!
4.		Identify what you will critique in these perspectives i.e. how to analyse/evaluate these perspectives. Critiques might consider ONE of the following: - how much evidence supports the two theories or interpretations or conclusions? - to what extent does the new evidence support the old assumption? - which theory or approach can better explain the phenomenon? - to what extent does efficacy outweigh limitations? - to what extent is this idea feasible? - or various others!
5.		Identify trustworthy sources that can help build your arguments, ideas and evidence around the critique of two perspectives. To help you, use the online library guide . It shows how to find quality sources efficiently and how to use appropriate tools that are specifically designed for source-finding.

* Here are some *examples* of topics and perspectives, but you should identify your own.

Topic: CRISPR. **Perspective 1:** long-term challenge. **Perspective 2:** short-term challenge.

Topic: Quantum computing. **Perspective 1:** scalability of software. **Perspective 2.** Scalability of hardware

Topic: Biodegradable plastics. **Perspective 1:** photodegradation. **Perspective 2.** Microbial degradation.

Topic: Origin of life on Earth. **Perspective 1:** panspermia theory. **Perspective 2.** Primordial soup theory.

Part 2: a screencast video reflection on your blog post

Purpose: show that you understand how your blog post demonstrates the assessed skills

Length: 3 - 4 minutes

Requirements: Your video reflection must screenshare/show your screen (we recommend using Zoom) so that we can see your blog post on the video screen while you talk. Highlight some of your best examples of the assessment skills in your blog post and orally explain why/how each example shows this skill or why you included it in this way.

- Do not create any visuals; just scroll through your own blog post on the screen as your visual.
- Do not use an AI voice; just talk yourself.
- Do not create a script; just talk naturally.

IMPORTANT:

- It is your responsibility to let your instructor know **in advance** if you have a problem or require special consideration.
- Late submissions for the assessment will be **marked down by one band across all assessed criteria**. Assignments submitted **more than 48 hours after the deadline will score a zero**.
- Written assignments outside the acceptable word range will be **marked down by one band across all assessment criteria**.
- Written assignments submitted without a reflection video will be **capped at "satisfactory" level**.
- Assignments with unethical use of sources or unethical use of AI or unethical plagiarism **may score a zero**.